# Examining the Effects of Precipitation on Flood Severity in a Small Watershed in Central Kentucky

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## INTRODUCTION

Flooding has become a significantly more common weather event in the state of Kentucky. According to the National Oceanic and Atmospheric Administration, from February 1st, 2024, to February 28th, 2025, 157 flood and 137 flash flood events were reported for a total of 294 flood-related events. Despite the recurrence of flooding events, weather forecasts and actual precipitation data are insufficient to elucidate or predict the locations of floods. To conduct a comprehensive investigation into how precipitation alone may impact flooding, we modeled a small watershed near Mundy's Landing with a series of storm scenarios.

## METHODS

To develop two-dimensional flood models utilizing the workflow of Swallom et al. (2025) and lidar-derived data Ison et al. (2025), precipitation data was downloaded from Western Kentucky University's Mesonet stations from the past five years (see Figures 1 and 2). The inverse-distance-weighting method was employed to estimate precipitation within the catchment area. This approach involved collecting data from the four nearest counties and interpolating the precipitation within the polygon (refer to Figure 2). ArcGIS Pro was utilized to generate 15-minute interval precipitation grids compatible with TUFLOW software, thereby facilitating the creation of a base model. Subsequently, additional storm events and temporal intervals, including hourly data for July 2021 and February 2023, were tested. A hypothetical storm for the year 2100 was generated by multiplying the July 2021 precipitation by 1.28 to approximate a projected precipitation increase of 7% per degree Celsius (Education, n.d.).

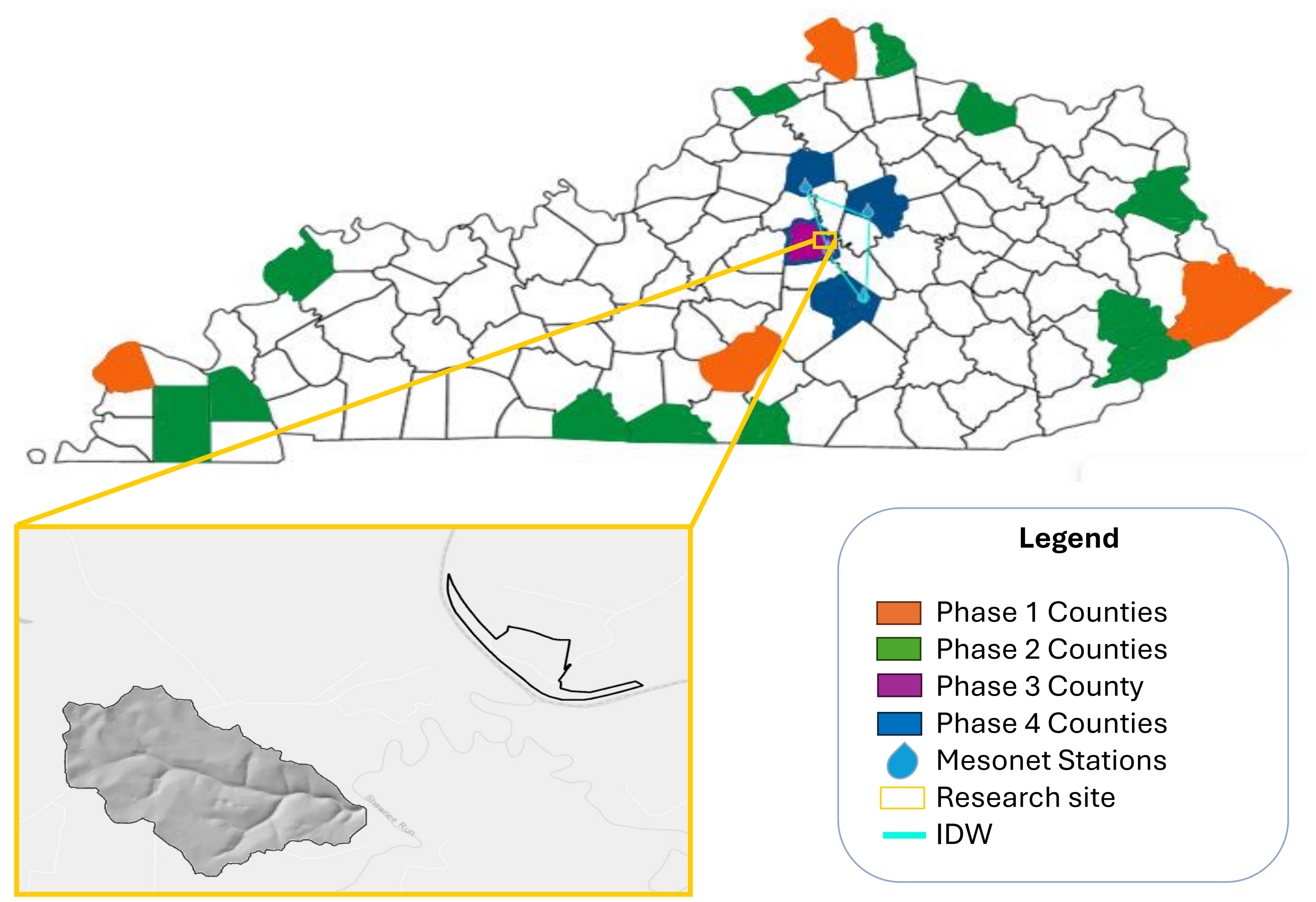


**Figure 1**: Map of the state of Kentucky featuring the various counties used for research. Phase 1 was used for looking at general trends in precipitation. Phase 2 was used for a close-up overview of weather patterns. Phase 3 was used to examine weather data close to research site. Phase 4 was used to get the IDW values in the catchment area/research site using the closest counties to Mundy's Landing.

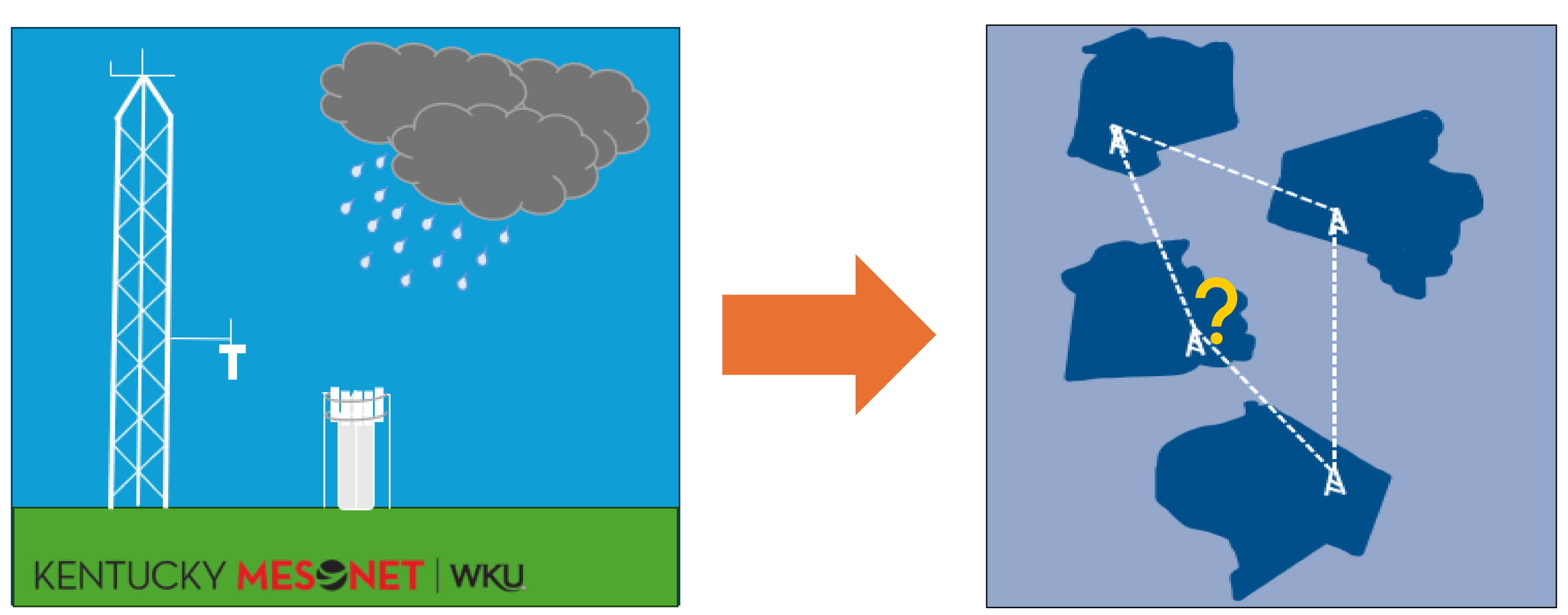


**Figure 2**: Depiction of precipitation data collection at WKU's Mesonet station. The precipitation data is used for the inverse distance weighting method (IDW), where precipitation amounts are taken from four points and used to find the precipitation rate in the catchment area, represented by the gold question mark.

## RESULTS

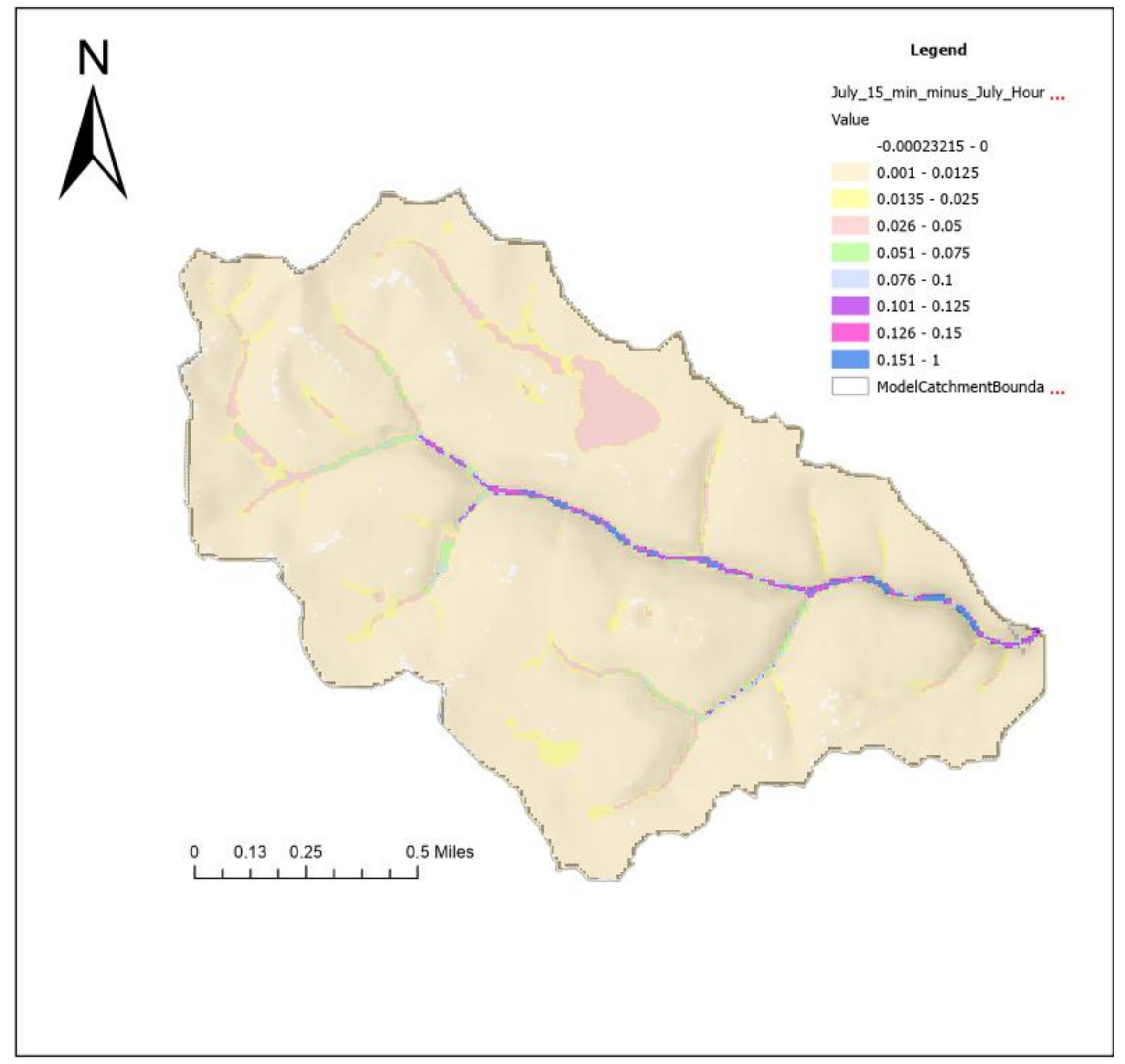


**Figure 3**: Difference map comparing flood heights between July 2021 with 15-minute data versus hourly. The map depicts a clear increase in modeled flood height due to an increased rate with the 15-minute data, not an increased overall precipitation.

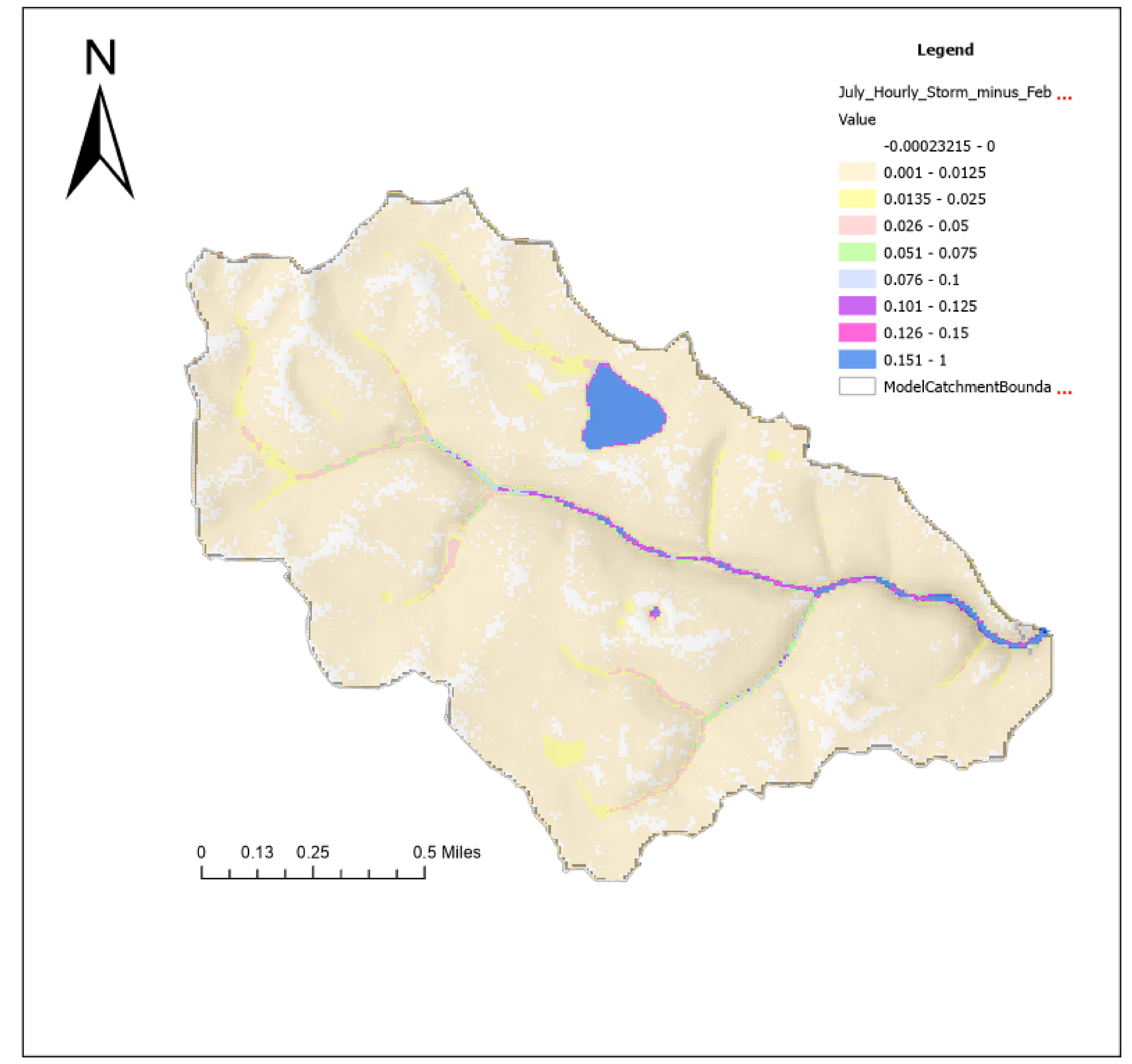


**Figure 4**: Difference map comparing flood heights between July 2021 and February 2023. The July storm had significantly higher flood heights compared to the February storm. This is caused by the brevity of the July storm compared to the drawn-out February storm. Future investigations could focus on seasonal weather patterns and temporal variations.

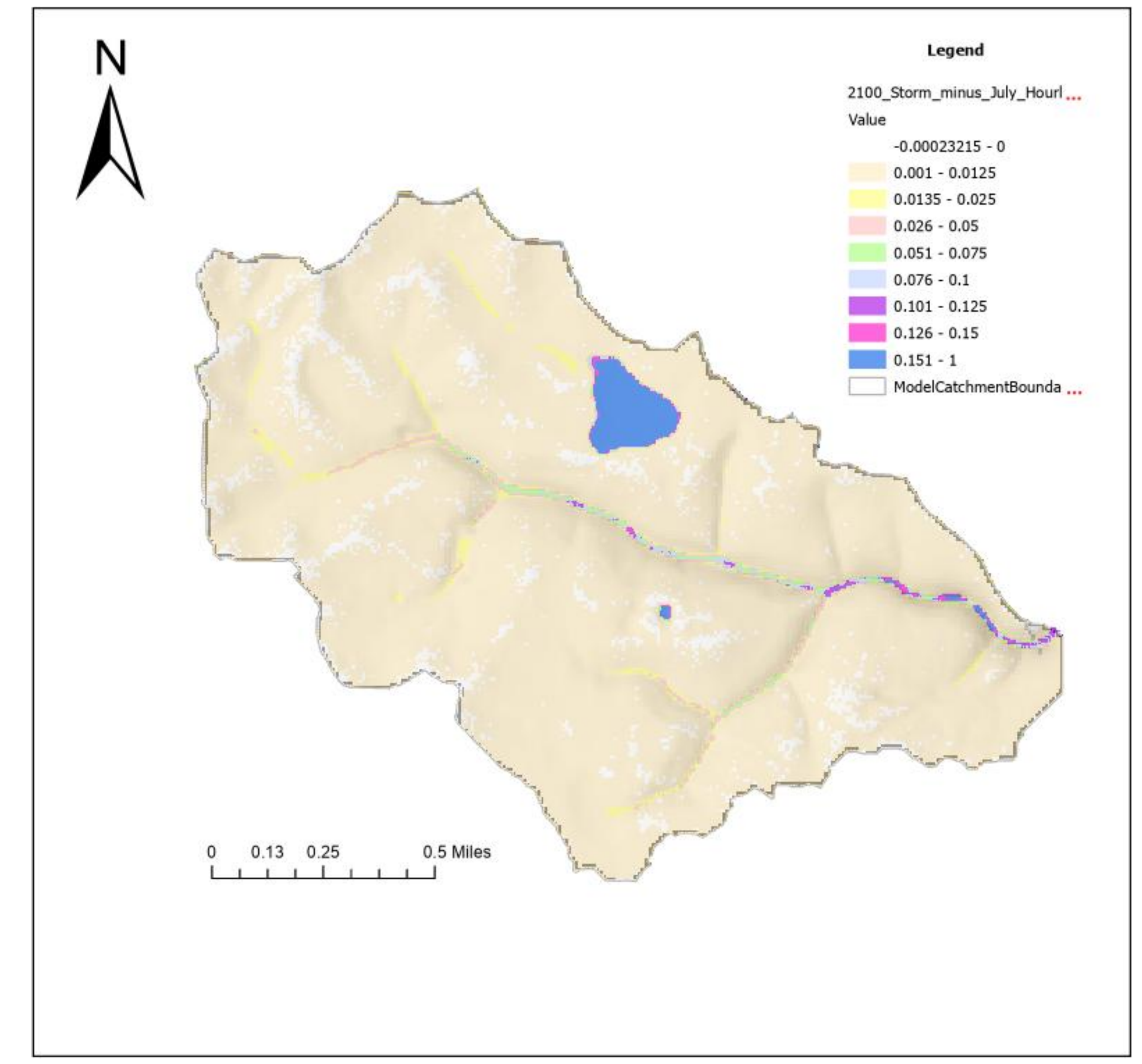


**Figure 5**: Difference map comparing flood heights between 2100 storm and the July hourly storm data. The predicted 2100 storm had significantly higher rates of precipitation, but flood heights between this storm and the 15-minute data were closely related. The predicted storm was created through projections made by University Corporation for Atmospheric Research (UCAR).

## CONCLUSION

**Figure 6**: This figure depicts how timing affects flooding. A short storm with heavy precipitation leads to flooding.

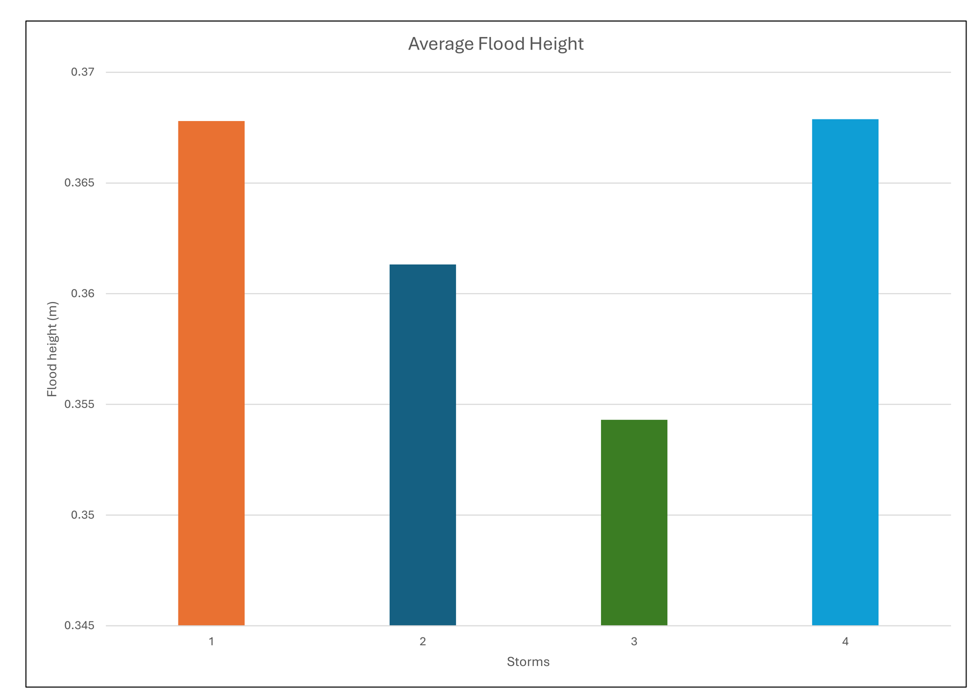


**Figure 7**: This bar chart the mean flood height for each model output. The February storm had the lowest flood height, while July 2021 and 2100 storm had the highest flood height.

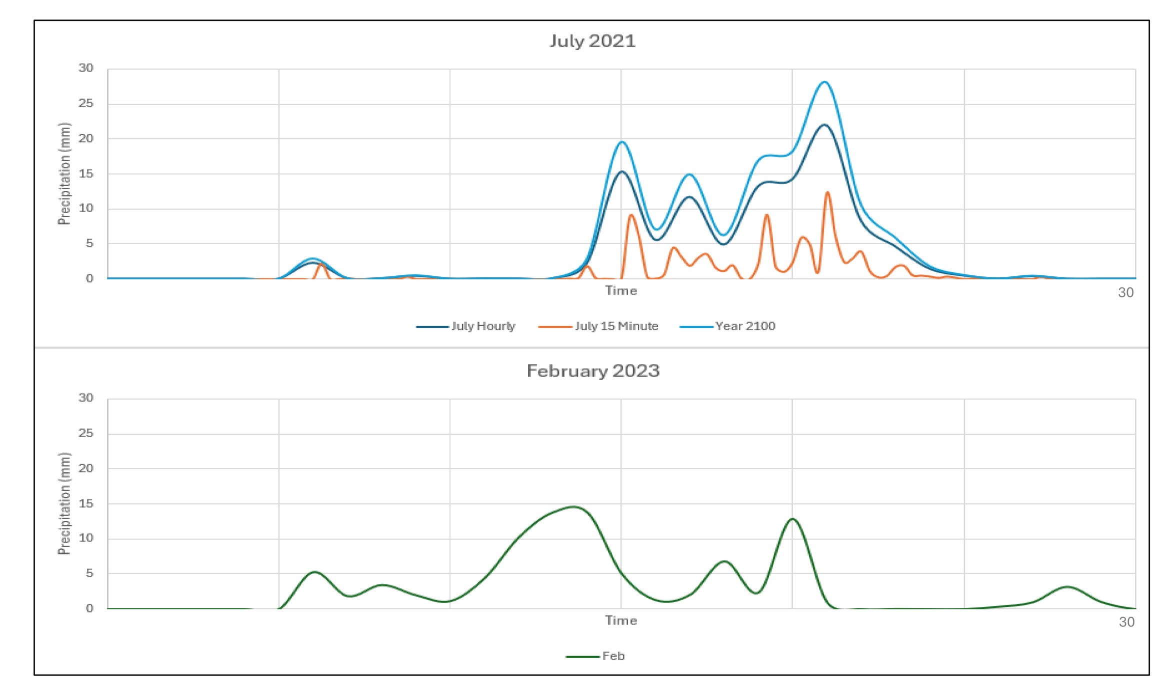


**Figure 8**: This figure depicts the comparison of precipitation distributions for all storms used in the catchment model. The 15-minute model is more accurate and shows more variability but takes longer to model.

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